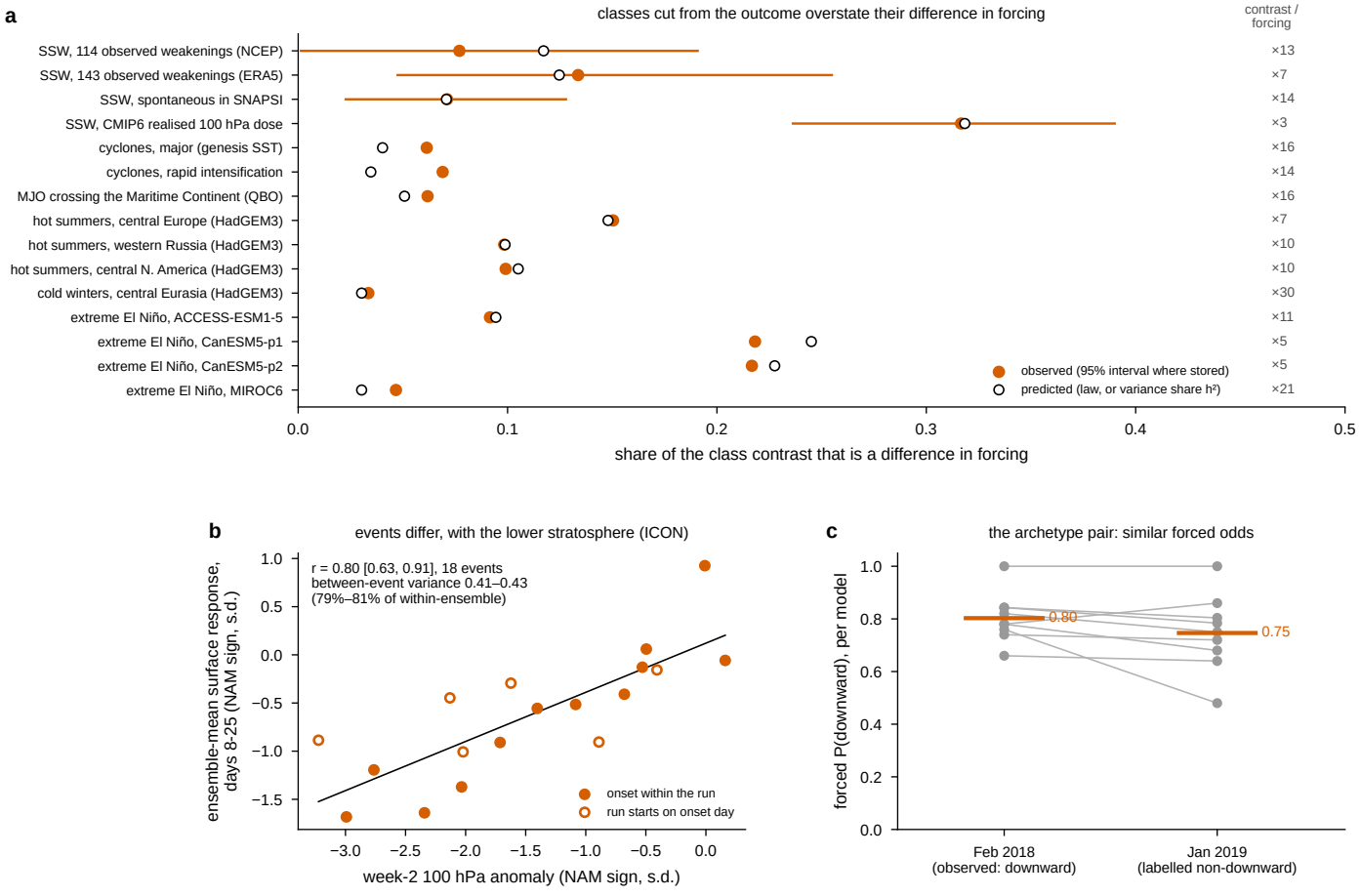
a
classes cut from the outcome overstate their difference in forcing
contrast / forcing
SSW, 114 observed weakenings (NCEP)
SSW, 143 observed weakenings (ERA5)
SSW, spontaneous in SNAPSI
SSW, CMIP6 realised 100 hPa dose
cyclones, major (genesis SST)
cyclones, rapid intensification
MJO crossing the Maritime Continent (QBO)
hot summers, central Europe (HadGEM3)
hot summers, western Russia (HadGEM3)
hot summers, central N. America (HadGEM3)
cold winters, central Eurasia (HadGEM3)
extreme El Niño, ACCESS-ESM1-5
extreme El Niño, CanESM5-p1
extreme El Niño, CanESM5-p2
extreme El Niño, MIROC6
×13
×7
×14
×3
×16
×14
×16
×7
×10
×10
×30
×11
×5
×5
×21
observed (95% interval where stored)
predicted (law, or variance share h²)
0.0
0.1
0.2
0.3
0.4
0.5
share of the class contrast that is a difference in forcing
b
events differ, with the lower stratosphere (ICON)
r = 0.80 [0.63, 0.91], 18 events
between-event variance 0.41–0.43
(79%–81% of within-ensemble)
ensemble-mean surface response, days 8-25 (NAM sign, s.d.)
1.0
0.5
0.0
−0.5
−1.0
−1.5
onset within the run
run starts on onset day
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
week-2 100 hPa anomaly (NAM sign, s.d.)
c
the archetype pair: similar forced odds
forced P(downward), per model
1.0
0.8
0.6
0.4
0.2
0.0
0.80
0.75
Feb 2018
(observed: downward)
Jan 2019
(labelled non-downward)